

Kansas State University
Carl and Melinda Helwig Department of Biological and Agricultural Engineering
920 N. Martin Luther King Jr. Drive, Manhattan, KS 66506

Dear Members of the Faculty Search Committee,

My work has always centered on one question: how do we build technologies that clean our environment without creating new problems? That commitment shapes my enthusiastic interest in the Assistant Professor position in the Carl and Melinda Helwig Department of Biological and Agricultural Engineering at Kansas State University (K-State). With a background that **bridges environmental and chemical engineering**, I am eager to contribute my expertise and passion to the esteemed community.

My research vision is to develop chemical-free advanced treatment and remediation processes that support safe water reuse and circular management of treatment residuals. I aim to integrate non-thermal plasma (NTP), micro-and nanobubble technologies, and nanomaterial-based methods to remove persistent contaminants from water and soil systems. Specifically, I propose to: (1) Design NTP systems coupled with micro- and nanobubbles for wastewater reclamation; (2) Design 3D-printed architected adsorbent materials for soil and water remediation with recoverable and regenerable purposes; and (3) Develop decentralized strategies for regenerating contaminant-laden adsorbents using plasma and microwave-based processes. Together, these modular, scalable approaches create a unified framework for advanced environmental treatment, resource recovery, and long-term resilience, aligned with K-State's commitment to sustainability, interdisciplinarity, and societal impact.

Currently, as a Postdoctoral Research Associate at the University of Maine, I am leading projects to design NTP systems for degrading PFAS to remediate contaminated soils (as a Co-PI, funded by Maine DACE) and regenerate PFAS-laden granular activated carbon. Additionally, I am serving as one of the lead team members in building *Maine's largest PFAS+ Research Center* at UMaine. Together, these efforts reflect my goal of creating scalable treatment technologies while connecting fundamental research with real-world applications. My research contributions have been recognized nationally and internationally. I was selected as a ***Trailblazers in Engineering (TBE) Fellow 2025*** by **Purdue University**, a highly competitive program designed to prepare outstanding future engineering faculty. I have also been honored as a ***40 Under 40 Rising Star in the Class of 2024*** by the American Academy of Environmental Engineers and Scientists (AAEES, USA), recognizing my contributions to environmental engineering research. **I have co-authored nine research grant applications as a Co-PI, seven of which have been funded.** I have published papers in top environmental engineering journals and presented my work at leading conferences.

At K-State, I am excited to collaborate with the Kansas Water Institute cohort, Helwig BAE faculty, and Environmental Engineering faculty to develop detection and mitigation approaches for traditional (nitrate and other nutrients, uranium) and emerging contaminants (PFAS, PPCPs, pesticides) in surface and groundwater, with interconnections to sustainable wastewater treatment for reuse to enable circular economies at interfaces with agriculture and municipal systems, including irrigation reuse. In the near future, I plan to pursue the NSF CAREER Award focused on the fundamental science of non-thermal

plasma in water treatment, along with proposals through the NSF CBET-1440 and CHE 6882. I will also seek opportunities through the DoD's SERDP and ESTCP initiatives, USDA's NIFA AFRI and ARS programs, and foundations, including Water Research Foundation (WRF), and Bill Melinda Gates Foundation. At the state level, I will pursue support from the Kansas Department of Health and Environment (KDHE), Kansas Water Office, Kansas Department of Agriculture's conservation cost-share programs, Kansas NSF EPSCoR seed funding opportunities, and the Kansas Water Institute/Kansas Water Resources Institute competitive research support through the USGS Water Resources Research Act program. With my grant-writing experience and strong publication record, I am confident in my ability to secure funding and launch my research program in your esteemed department.

In addition to my research, I have a deep commitment to teaching and mentoring the next generation of engineers and scientists. **As a first-generation graduate in my family**, I have navigated various challenges that have profoundly shaped my perspective on education, making empathy and justice integral to my approach as an educator and researcher. This has driven me to create a classroom and lab environment that is not only empathetic but also empowering for students from all backgrounds. During the Spring 2024 and 2025 semesters, I had the privilege of teaching the "Wastewater Process Design" course with lab as a sole instructor at the University of Maine. My approach combined traditional lectures with flipped classroom strategies, active learning, and visits to wastewater treatment plants, which was well received by students. In both semesters, I received over 90% positive feedback (Excellent and Very Good), with students noting my accessibility, commitment to their success, and supportive and engaging teaching style. At K-State, given my teaching and research background, I look forward to contributing to courses such as, (1) ENVE 363: Environmental Engineering Fundamentals, (2) CE 565: Water and Wastewater Engineering, (3) CE 762 - Water Treatment Processes, and (4) CE 663 - Unit Operations and Processes in Environmental Engineering, as well as developing new courses on (1) Environmental Nanotechnology: Applications & Implications, (2) Sustainable Water Reuse and Circular Water Systems, and (3) Advanced Water Treatment Process. I plan to establish a summer program for high school students from underrepresented communities, offering Learn-by-Doing experiences that build lifelong connections and inspire students to embrace their unlimited possibilities at K-State.

I am confident in contributing to K-State's academic environment through my commitment to advancing sustainable environmental remediation processes, fostering an immersive classroom, and interdisciplinary collaboration. I look forward to discussing how my research, teaching, and mentorship expertise can contribute to the vision of the Carl and Melinda Helwig Department of Biological and Agricultural Engineering at K-State. I have enclosed my CV with references, research, and teaching statements for your review. Thank you for considering my application.

Sincerely,

Manisha Choudhary

Manisha Choudhary, Ph.D.

Postdoctoral Research Associate